linnest

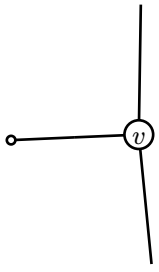
a typst interface to the `linnet` crate

`linnet` is a rust crate that provides a graph data structure and many graph algorithms, and crucially for this package, multiple layout algorithms, as well as a dot parser (using the `dot_parser` crate).

Contents

Linnet

The `linnet` crate that this package wraps around, is built around the half-edge graph data structure. This means that instead of a graph being represented as a set of nodes and edges, it is represented as a set of half-edges H, and a set of vertices V. The graph structure is then encoded through two maps. The first map, $\partial : H \longrightarrow V$ maps each half-edge to its corresponding vertex. The preimage of any vertex v is the set of half-edges that map to it, called the crown of v. The second map, $\iota : H \longrightarrow H$, is an involution that *glues* half edges together to form edges. If a half-edge is glued to itself, we call that an external half-edge. This means that linnet graphs are strictly more capable than normal edge and vertex graphs.



A side effect of natively supporting half-edges, is that subgraphs can be more granular, as we can encode them as sets of half-edges. This naturally supports vertex induced subgraphs (the union of the crowns of a set of vertices).

Linnest

Linnest is the typst plugin + API to integrate with linnet and access some of its capabilities. This means layouting algorithms, and dot parsing (all through wasm, so no dependencies/ external tooling required) along with a selection of graph algorithms (that can be expanded upon as needed).

Graphs can be constructed in two ways, parsing from a dot string using `parse`:

```
#let g = parse("digraph { a -> b }")
```

or building from edges and nodes using `build`, with a fletcher inspired syntax:

```
#let g = build({
  node(<a>, label: [$v$])
  node(<b>)
  edge(source(<a>), <a-b>, sink(<b>), label: [e])
  edge(source(<a>), <in-a1>, label: [e])
  edge(source(<a>), <in-a2>, label: [e])
})
```

In either case, `type(g)=dictionary`, because graph values are Typst dictionaries that wrap an archived linnet graph plus native Typst data. Rust owns topology, layout state, statement metadata, and internal opaque payload bytes; user data captured from Typst stays in Typst and is merged back into query records.

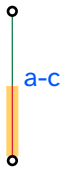
The main usecase however is of course to automatically place the nodes and edges on a canvas, using the `layout` function.

- `graph` for construction, parsing, inspection, joins, and graph algorithms.
- `subgraph` for subgraph object construction and inspection.
- `layout` for the separate layout pass.
- `draw` for rendering a laid-out graph object with CeTZ.
- `physics` for reusable particle-line edge styles.

Minimal Build Example

```
#import "../src/lib.typ": draw, graph, layout, subgraph
#import graph: build, dot, edge, edges, node, nodes, parse, sink, source

#let g = build({
  node(<a>)
  node(<c>)
  edge(
    source(<a>, compass: "e"),
    <a-c>,
    sink(<c>, compass: "w"),
    label: [a-c],
    statements: (
      color: "0055ff",
      source-color: "d72638",
      sink-color: "1b7f4c",
    ),
  )
},
  name: "demo",
)
#let g = layout(g)
#let east = subgraph.compass(g, "e")
#let edge-records = edges(g, subgraph: east)
#let dot-text = dot(g)
#let edge-label(edge) = text(fill: rgb("#" + edge.color))[#edge.label]
#let source-style(edge) = (stroke: rgb("#" + edge.source-color) + 0.5pt)
#let sink-style(edge) = (stroke: rgb("#" + edge.sink-color) + 0.5pt)
#draw(
  g,
  subgraph: east,
  edge-label: edge-label,
  edge-label-style: (anchor: "south"),
  source-style: source-style,
  sink-style: sink-style,
)
```



Graph Objects

Graph objects are Typst dictionaries wrapping archived Rust graph bytes plus native Typst data arrays for graph, node, edge, source, and sink data. The Rust graph may also carry an internal opaque payload, but that payload is used only by the Typst wrapper and is not exposed in public records. Build or parse graph objects with `graph`, transform graph objects with `layout`, and pass objects back to `graph` or `subgraph` for inspection. Subgraph objects are still opaque zero-copy values.

- `graph.parse(input)` parses one or more DOT digraphs and returns an array of graph objects. Its `eval-graph-fields`, `eval-node-fields`, `eval-edge-fields`, `eval-source-fields`, and `eval-sink-fields` arguments are convenience arguments for `graph.eval-fields`.
- `graph.build(..)` constructs one graph object from a stream of node and edge items.
- `graph.map(graph, ..)` maps graph, node, edge, source, and sink records to new native data without changing topology.
- `graph.node-data(graph, <name>)` and `graph.edge-data(graph, <name>)` return one named node or edge data.
- `graph.update-node-data(graph, <name>, data)` and `graph.update-edge-data(graph, <name>, data)` update one named node or edge data. Direct replacements and callbacks run in Typst with `(data, record)`.
- `graph.eval-fields(graph, ..)` evaluates selected record fields into data entries. It works on parsed and built graph objects.
- `node(..)` returns a node item.
- `source(..)` and `sink(..)` return half-edge endpoints.
- `edge(..)` returns an edge item built from source/sink endpoints and an optional edge name.
- `layout(graph, ..)` runs layout as an explicit second step. Its settings are named parameters so calls stay descriptive and Tidy can document each field.
- `draw(graph, ..)` draws a laid-out graph object with CeTZ.
- `dot(graph)` returns a DOT string for inspection or export.

Graph Specs

The Typst construction API is half-edge first: create node items, create source and sink half-edge endpoints, then pass edge items to `graph.build`. `graph.build` accepts both comma-separated items and ordinary Typst code blocks. Typst labels such as `<a>`, `<h1>`, and `<e1>` are API names; node and edge names are emitted back from `nodes(g)` and `edges(g)` as Typst labels. They are resolved before the wire format is sent to the Rust plugin. Numeric `id` arguments choose graph indexes or ordering: on nodes, `id` fixes the resulting node index and must be unique and in bounds. On nodes, edges, sources, and sinks, extra named arguments are captured as opaque Typst data fields: `edge(source(<a>, style: physics.source-stroke()), sink(<b>), particle: "g")` stores `(style: ..)` in the source data and `(particle: "g")` in the edge data. These user data values are not sent through the Rust plugin boundary. Typst sends an internal opaque payload with correlation data, asks Rust to resolve the graph indexes, then stores user data in arrays at those resolved graph, node, edge, and half-edge ids. `graph.info`, `graph.nodes`, and `graph.edges` merge those native values into the returned records. A captured `label` data field on nodes and edges is display content used by the default drawing style; use `statements: (label: "...")` when a flat metadata label string is needed. Statements are flat metadata used by DOT; they cannot nest. Values are scalar strings/numbers/booleans. Use data fields for structured Typst data or content.

`graph.build` and `graph.parse` also accept `default-node-data`, `default-edge-data`, `default-source-data`, and `default-sink-data`. These defaults are merged into the corresponding data; captured data fields on nodes, edges, sources, and sinks override the defaults. For drawing, the physics helpers read `data.style` on source and sink half-edges and `data.display-label`/`data.label` on edges:

```
#let g = build({
  node(<a>, label: [a])
  node(<b>, label: [b])
  edge(
    source(<a>, style: physics.source-stroke(c: red)),
    <e>,
    sink(<b>, style: physics.sink-stroke(c: blue)),
    label: [$p$],
    particle: "g",
  )
},
  default-edge-data: (kind: "propagator"),
)
#let callbacks = physics.style()
#draw(
  layout(g),
  source-style: callbacks.source-style,
  sink-style: callbacks.sink-style,
  edge-label: callbacks.edge-label,
)
```

When parsing DOT, the same native data arrays can be filled from selected string fields. Rust parses DOT into topology and statement metadata; Typst applies defaults and evaluates selected fields afterward. The selected fields are evaluated with the record's merged `fields` dictionary in scope. Since node names are labels, use `#str(name)` when a name should become visible text:

```
#let g = parse(
  "digraph g { a [label=\"A\"]; a -> b [label=\"$p$\", source=\"out\", sink=\"in\"] }",
  eval-node-fields: ("label",),
  eval-edge-fields: ("label",),
  eval-source-fields: ("statement",),
  eval-sink-fields: ("statement",),
).first()
#nodes(g).first().data.label
#edges(g).first().data.label
```

The same transform can run after construction. This is useful for global edge statements that should apply to every edge while still seeing local edge fields:

```
#let g = build({
  node(<a>)
  node(<b>)
  edge(source(<a>), sink(<b>), statements: (mom: "p"))
}, default-edge-statements: (display-label: "$#mom$"))
#let g = graph.eval-fields(g, eval-edge-fields: ("display-label",))
#edges(g).first().data.at("display-label")
```

```
#let g = build({
  node(<a>)
  node(<c>)
  edge(source(<a>), <e1>, sink(<c>))
})
```

Named nodes and edges can be updated after construction without scanning in the caller. The update replaces the native data, or it can be a callback receiving `(data, record)`:

```
#let g = build({
  node(<a>)
  node(<c>)
  edge(source(<a>), <e1>, sink(<c>))
})
#let g = update-node-data(g, <a>, (label: [A]))
#let g = update-edge-data(g, <e1>, (data, edge) => (
  label: [$p$],
  source: edge.source.node,
))
```

```
#node-data(g, <a>).label
#edge-data(g, <e1>).label
```

`source(..)` and `sink(..)` accept a node reference plus optional `name`, `id`, `statement`, and `compass`. `name` is a Typst label name for the half-edge; `id` is numeric:

```
edge(source(<a>, name: <h1>, id: 0), <e1>, sink(<c>, id: 2), label: [a-c])
```

One half-edge creates an external edge. The side is determined by the constructor, so there is no public `flow` argument:

```
edge(<incoming>, sink(<a>))
edge(source(<c>), <outgoing>)
```

`graph.build` does not interpolate statement strings on the Rust side. Use `graph.eval-fields` or `graph.map` when a default statement should turn into Typst content or structured data data. The evaluation scope includes the record's merged fields, so default edge statements can still refer to local edge fields:

```
#let g = build({
  node(<a>)
  node(<c>)
  edge(
    source(<a>),
    <a-c>,
    sink(<c>),
    label: [a-c],
    statements: (color: "0055ff", label: "a-c"),
  )
},
  default-edge-statements: (
    color: "000000",
    display-label: "$#label$",
  ),
)
#let g = graph.eval-fields(g, eval-edge-fields: ("display-label",))
```

Placements

`graph.pos` creates a first-class placement. The default `mode: "pin"` turns a coordinate into a fixed layout constraint and a drawable position. Use `mode: "start"` when the coordinate should only seed the layout:

```
#let g = build({
  node(<a>, pos: graph.pos(x: -2, y: 0))
  node(<c>, pos: graph.pos(ref: <a>, dx: 4, dy: 0))
  edge(source(<a>), <a-c>, sink(<c>), pos: graph.pos(x: 0, y: 1.2))
})
```

`graph.group` links one coordinate across several nodes or edge control points. A `side` of `"+"` keeps the coordinate positive, and `"-"` keeps it negative. GammaLoop external-edge columns use this to keep incoming and outgoing external legs on opposite sides while pairing rows by a shared `y` group:

```
#let edge-items = (
  edge(
    source(<right-ext>),
    sink(<center>),
    pos: graph.pos(x: graph.group("right", side: "+"), y: graph.group("edgee0")),
  ),
  edge(
    source(<center>),
    sink(<left-ext>),
    pos: graph.pos(x: graph.group("left", side: "-"), y: graph.group("edgee0")),
  ),
)
```

Raw DOT input uses the same placement model through the `pos` attribute. The standard Graphviz subset is preserved: `pos="x,y"` is a starting coordinate and `pos="x,y!"` is pinned. Linnest extends the value with

explicit id references and axis entries. In axis entries, `!` belongs to that axis: numeric entries without `!` are starting coordinates, numeric entries with `!` are fixed constraints, and grouped entries must use `!` because groups are constraints.

```
digraph {
  a [id=0 pos="0,0!"]
  b [id=1 pos="ref(node:0)+4,0!"]
  a -> b [id=0 pos="ref(node:1)+0,1!"]
  b -> c [id=1 pos="ref(edge:0)+1,0!"]
  c [id=2 pos="x:2!"]
  d [id=3 pos="x:2!,y:1"]
  ext -> a [id=2 pos="x:@-left!,y:@edge0!"]
}
```

The DOT grammar is intentionally id-based: `ref(node:0)` uses a node `id` and `ref(edge:0)` uses an edge `id`. Bare names and implicit edge order are not placement references. The `@` syntax denotes grouped coordinate constraints; `@+name` and `@-name` keep the grouped coordinate on the positive or negative side respectively.

```
pos="x,y"                   // start x and y
pos="x,y!"                  // pin x and y
pos="x:<coord>"             // start numeric x
pos="x:<coord>!"            // pin x
pos="y:<coord>!"            // pin y
pos="x:<coord>!,y:<coord>"  // pin x, start numeric y
pos="x:<coord>,y:<coord>!"  // start numeric x, pin y
```

Draw styling is Typst-native. Pass dictionaries or callbacks to `draw`; edge callbacks receive the merged `scope`, edge statements, source/sink half-edge records, and the edge index:

```
#let edge-label(edge) = text(fill: rgb("#" + edge.color))[#edge.display-label]
#let source-style(edge) = (stroke: red + 0.5pt)
#let sink-style(edge) = (stroke: blue + 0.5pt)
#draw(layout(g), edge-label: edge-label, source-style: source-style, sink-style: sink-style)
```

Edge labels are drawn with CeTZ content at the layout label position. Use `edge-label-style: (anchor: "south")`, or an edge-data callback returning a style dictionary, to choose which point of the label is anchored there.

Marks can follow graph orientation as a first-class draw option. Put the same mark layer on both halves and set `mark-orientation: "edge"`; default-oriented edges mark the source half, reversed edges mark the sink half with the marker flipped, and undirected edges suppress the mark.

```
#let oriented-arrow = (
  stroke: black + 0.7pt,
  mark: (end: (symbol: ">", fill: black, anchor: "center", shorten-to: auto), scale: 0.75),
  mark-position: "center-if-dangling",
  mark-orientation: "edge",
)
#draw(layout(g), source-style: oriented-arrow, sink-style: oriented-arrow)
```

Physics Edge Styles

`physics.style(..)` returns `source-style`, `sink-style`, and `edge-label` callbacks for `draw`. The default source/sink strokes use darker/lighter halves to encode the graph source/sink split. Fermion map entries marked with `fermion-flow` receive one particle-flow arrow on the main edge using `mark-orientation: "edge"`; paired edges follow `edge.orientation`, dangling half edges place the arrow in the middle of the visible half edge, and undirected edges omit the arrow. This is independent of momentum arrows.

Set `momentum-arrows: true` to draw centered parallel arrow decorations while the main edge remains connected to the nodes. The arrows flow from source to sink independently of the DOT `dir`/`orientation` value; there is one momentum marker per edge, even though the base edge is internally styled as source and sink halves. On paired edges the marker lives on the sink half; on dangling edges it lives on the existing source or sink half-edge. The decoration defaults to a plain black `0.55pt` stroke and a scaled CeTZ

`"barbed"` arrowhead, so it does not inherit the base particle stroke. The arrow length is capped by both `momentum-arrow-length` and `momentum-arrow-ratio`, so the shorter one wins. When the layout provides an edge-label position, the arrow offset is signed so the momentum marker is drawn on the same side of the edge as that label. Use `momentum-arrow-offset` to set the normal displacement. `momentum-arrow-mark: auto` uses the default barbed marker, `momentum-arrow-mark: none` draws only the momentum line, and any other value overrides the CeTZ mark style, for example `"stealth"` or `(end: "barbed", scale: 1.6)`. Only the `end` mark is kept so source/sink splitting cannot create a second momentum head.

Optional labels can be built from edge metadata with `show-edge-index`, `show-half-edge-index`, `show-particle`, and, for explicit momentum fields, `show-momentum`. `show-half-edge-index` only emits a value for dangling half edges.

```
#import "../src/lib.typ": draw, graph, layout, physics
#import graph: build, dot, edge, edges, node, nodes, parse, sink, source

#let g = build({
  node(<a>)
  node(<c>)
  edge(
    source(<a>),
    <fermion-main>,
    sink(<c>),
    id: 7,
    particle: "fermion",
  )
  edge(
    source(<c>),
    <fermion-out>,
    id: 8,
    particle: "fermion",
  )
},
  name: "physics",
)
#let callbacks = physics.style(
  momentum-arrows: true,
  show-edge-index: true,
  show-particle: true,
)
#draw(
  layout(g),
  source-style: callbacks.source-style,
  sink-style: callbacks.sink-style,
  edge-label: callbacks.edge-label,
)
```

Set `edge-offset` on `draw`, or `offset` in a `source-style`/`sink-style` dictionary, to draw a fitted parallel path. Style callbacks may also return an array of dictionaries; the layers are drawn in order on the same graph edge, so parallel strokes do not require duplicate graph edges.

```
#let base-style(edge) = (
  stroke: (paint: gray, thickness: 0.7pt, cap: "round"),
)
#let offset-style(edge) = (
  offset: 0.18,
  length: 1.4,
  ratio: 0.5,
  resolve-length: "min",
  stroke: (paint: rgb("#2f6f4e"), thickness: 1pt, cap: "round"),
)
#let source-style(edge) = (base-style(edge), offset-style(edge))
#let sink-style(edge) = (base-style(edge), offset-style(edge) + (mark: (end: ">")))
```

The parallel path is applied to the base edge geometry before patterns and other decorations; node outsets then trim the shifted path, so shifted paths still start and end outside fitted node circles. Add `edge-length` or `length` to center-trim the shifted path to a fixed arc length, and add `edge-ratio` or `ratio` to cap it by a fraction of the base edge length. `edge-resolve-length`
`resolve-length` decides how to combine both limits `"min"`/`"shorter"` (default), `"max"`/`"longer"`, `"length"`/`"fixed"`, `"ratio"`/`"relative"`, `"none"`/`"full"`, or a function receiving `(base-length, length, ratio)`. Set `offset-side: "label"` on an offset layer to choose the sign of `offset` so the layer is drawn on the same side of the curve as the edge label.

Data defaults are also the global styling hook for all sources, sinks, nodes, and edges. More specific data can be added with captured named arguments on node, edge, source, and sink items, or by running `graph.map`/`graph.eval-fields` after construction. This keeps evaluated Typst values in the native data channel instead of adding renderer-specific eval fields to the Rust topology spec:

```
#let g = build({
  node(<a>)
  node(<c>)
  edge(
    source(<a>),
    <a-c>,
    sink(<c>),
    label: [a-c],
    kind: "highlight",
  )
},
  name: "demo",
  default-source-data: (style: (stroke: red + 0.5pt)),
  default-sink-data: (style: (stroke: blue + 0.5pt)),
)
#let callbacks = physics.style()
#draw(layout(g), source-style: callbacks.source-style, sink-style: callbacks.sink-style)
```

`graph.build` also accepts comma-separated items:

```
#let g = build(
  node(<a>),
  node(<b>),
  edge(
    source(<a>, compass: "e"),
    <ab>,
    sink(<b>, compass: "w"),
    label: [ab],
    statements: (color: "0055ff", label: "ab"),
  ),
  name: "demo",
  statements: (full_num: "x + y"),
  default-node-statements: (shape: "circle"),
  default-edge-statements: (
    color: "000000",
    display-label: "$#label$",
  ),
)
```

Graph Queries

`graph.info(g)` returns graph metadata. `nodes(g)` returns node records, and `edges(g)` returns edge records. Node and edge record `name` values are Typst labels when present. Pass `subgraph: sg` to filter nodes or edges by an subgraph object.

`graph.join(left, right, key: "statement")` joins matching dangling half edges. The key is read from half-edge statements or numeric ids and can be `"statement"`, `"compass"`, or `"id"`.

`graph.cycles(g)` returns subgraph objects for a cycle basis. `graph.forests(g)` returns subgraph objects for spanning forests.

Layout Model

`layout` starts from a traversal-tree placement, then optimizes the positions of graph nodes and edge control points. The initial tree spacing is

$$L = \lambda \sqrt{\frac{WH}{\max(n, 1)}}$$

,

with horizontal spacing $\tau_x L$ and vertical spacing $\tau_y L$. Here λ is `length-scale`, W is `viewport-w`, H is `viewport-h`, τ_x is `tree-dx`, and τ_y is `tree-dy`. These fields set the geometry scale for both layout modes.

For deterministic, non-iterative placement, use `layout-algo: "tree"` or `layout-algo: "dot"`. `"tree"` places a traversal forest by levels. `"dot"` uses a directed layered placement for acyclic inputs, falling back to tree placement when the selected edges contain a cycle. Both modes also work on a subgraph:

```
#let g = parse("digraph partial { a -> b; b -> c; c -> d; d -> a }").at(0)
#let tree = graph.forests(g).at(0)
#let g = layout(g, layout-algo: "tree", subgraph: tree)
#draw(g)
```

Only nodes touched by the selected subgraph are placed. Edge control points are then resolved from the current node positions, so edges outside the selected subgraph are drawn as straight lines unless their own position constraints say otherwise.

`layout-algo: "force"` and `layout-algo: "anneal"` also accept `subgraph`. For these iterative modes, nodes and edge control points outside the selected subgraph stay fixed and act as boundary points while the selected subgraph is optimized.

Set `layout-nodes: "fixed"` to keep every node at its current `pos` for this layout pass and move only edge control points. With `subgraph`, only edges in the selected subgraph are moved; all other edge control points keep their current positions. The fixed-node policy is temporary; the returned graph stores the resulting coordinates as `pos`, but it does not turn them into persistent `pin` constraints. This is useful after one node-placement pass when a later pass should route or relax selected edges without disturbing the node layout:

```
#let g = parse("digraph partial { a [pos=\"0,0\"]; b [pos=\"4,0\"]; c [pos=\"8,0\"]; a -> b; b ->
c }").at(0)
#let first = subgraph.bits(g, (true, true, false, false))
#let g = layout(g, layout-algo: "tree", layout-nodes: "fixed", subgraph: first)
#draw(g)
```

The shared spring/charge model uses the following coefficients:

$$c_{\mathrm{vv}} = \beta L^2$$

$$c_{\mathrm{ev}} = \beta \gamma_{\mathrm{ev}} L^2$$

$$c_{\mathrm{ee}} = \beta \gamma_{\mathrm{ee}} L^2$$

$$c_{\mathrm{center}} = \beta g_{\mathrm{center}} L^2$$

$$c_{\mathrm{dangling}} = \beta \gamma_{\mathrm{dangling}} L^2$$

The Typst parameter names are `beta` for β, `gamma-ev` for γ_{ev}, `gamma-ee` for γ_{ee}, `g-center` for g_{center}, and `gamma-dangling` for $\gamma_{\mathrm{dangling}}$. The spring stiffness k is `k-spring`, and the softening constant ε is `eps`.

In `layout-algo: "anneal"`, linnest minimizes an energy:

$$E = \sum_{i<j} \frac{1}{2} \frac{c_{\mathrm{vv}}}{d(v_i, v_j) + \varepsilon} + \sum_{i,e} \frac{c_{\mathrm{ev}}}{d(v_i, e) + \varepsilon} + \sum_{(v,e)\ \mathrm{incident}} \frac{1}{2} k(\ell_e - d(v, e))^2$$

$$+ \sum_{\mathrm{local\ edge\ pairs}} \frac{1}{2} \frac{c_{\mathrm{ee}}}{d(e_i, e_j) + \varepsilon} + \sum_{\mathrm{dangling\ pairs}} \frac{1}{2} \frac{c_{\mathrm{dangling}}}{d(e_i, e_j) + \varepsilon} + \sum_i \frac{1}{2} \frac{c_{\mathrm{center}}}{d(v_i, 0) + \varepsilon} + p_{\mathrm{cross}} N_{\mathrm{cross}}$$

.

Here p_{cross} is `crossing-penalty` and N_{cross} is the number of detected edge crossings. `temp`, `step`, `seed`, `steps`, `epochs`, `cool`, `accept-floor`, `step-shrink`, and `incremental-energy` belong to this simulated annealing mode. `crossing-penalty` is also anneal-only; force mode does not currently add a crossing force.

In `layout-algo: "force"`, linnest applies the direct forces corresponding to the same vertex-vertex, edge-vertex, incidence spring, local edge-edge, dangling-edge, and center terms. `step` is the integration step, `delta` clamps per-step movement, `steps` and `epochs` set the iteration budget, `cool` shrinks the step after each epoch, and `early-tol` stops when movement is small. `z-spring` and `z-spring-growth` are force-only helpers: the integrator gives points temporary z coordinates to break overlaps and pulls them back toward the 2D plane.

`directional-force` is applied in both modes as an extra bias derived from pin/port direction constraints.

After either graph layout mode, labels are relaxed separately. If L_l is the label target distance and q_l is the label repulsion strength, then $L_l = \alpha_l L$ and $q_l = \beta_l L^2$. The Typst names are `label-length-scale` for α_l and `label-charge` for β_l. `label-spring` is the spring constant pulling each label toward its target. `label-layout: "normal"` uses a perpendicular offset target. With `label-layout: "dangling-tangent"`, paired edges still use that perpendicular target, but dangling half-edge labels are offset along the edge direction away from the attached node. With `label-layout: "fixed-length"`, the label remains at distance L_l from the edge point and only rotates around it under repulsive forces. `label-steps`, `label-step`, `label-early-tol`, and `label-max-delta-scale` control the label relaxation iteration.

Subgraphs

Subgraph objects are opaque zero-copy values.

- `subgraph.label(g, label)` constructs a subgraph from a base62 label.
- `subgraph.bits(g, bits)` constructs a subgraph from a boolean hedge array.
- `subgraph.compass(g, compass)` selects half edges with a DOT compass point.
- `subgraph.to-label(sg)` returns the base62 label.
- `subgraph.hedges(sg)` returns included hedge indices.
- `subgraph.contains(sg, hedge)` tests hedge membership.

Generated Reference

graph

- build()
- parse()
- node()
- source()
- sink()
- edge()
- group()
- pos()
- map()
- eval-fields()
- info()
- dot()
- nodes()
- edges()
- node-data()
- edge-data()
- update-node-data()
- update-edge-data()
- join()
- cycles()
- forests()

build

Build one graph object from a stream of node and edge items.

Use `node()`, `source()`, `sink()`, and `edge()` to create graph items. Positional items may be passed as comma-separated arguments or yielded from a Typst code block.

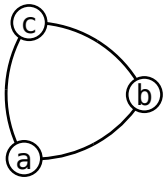
Parameters

```
build(
  ..items: array,
  name: none string,
  data: any,
  statements: dictionary,
  default-node-statements: dictionary,
  default-edge-statements: dictionary,
  default-node-data: any,
  default-edge-data: any,
  default-source-data: any,
  default-sink-data: any
) -> dictionary
```

..items `array`

Node and edge items returned by `node()` and `edge()`. The best way to use these is to use a code scope, so that the edges and nodes append each other:

```
#let g = build({
  node(<a>, label: [a])
  node(<b>, label: [b])
  node(<c>, label: [c])
  edge(source(<a>), sink(<b>))
  edge(source(<b>), sink(<c>))
  edge(source(<a>), sink(<c>))
})
```



name `none` or `string`

Graph name.

```
#let g = build(name: "My Graph", {
})
#info(g).name
```

```
My Graph
```

Default: `none`

data `any`

Native Typst graph data.

```
#let g = build(data: (a:(b:(1, ))), {
})
#info(g).data
```

```
(a: (b: (1,)))
```

Default: `none`

statements `dictionary`

Flat graph statements, that get turned into a string to string dictionary in rust. Used by DOT parsing; values cannot nest.

```
#let g = build(statements: (a:1), {
})
#info(g).global-statements
```

```
(a: "1")
```

Default: `(:)`

default-node-statements `dictionary`

Flat default node statements. Used by DOT; values cannot nest.

Default: `(:)`

default-edge-statements `dictionary`

Flat default edge statements. Used by DOT parsing; values cannot nest. These are applied/delegated to all edges.

```
#let g = build(default-edge-statements: (a:1),
{
   node(<a>)
   edge(source(<a>))
   edge(sink(<a>))
})
#edges(g).map(e=>e.statements)
```

```
((a: "1"), (a: "1"))
```

Default: `(:)`

default-node-data `any`

Default data merged into every node data. Captured node data fields override it.

```
#let g = build(default-node-data: (a:1), {
   node(<a>)
   node(<b>,a:2)
   node(<c>,b:(a:1))
})
#nodes(g).map(n=>n.data)
```

```
((a: 1), (a: 2), (a: 1, b: (a: 1)))
```

Default: `none`

default-edge-data `any`

Default data merged into every edge data. Captured edge data fields override it.

```
#let g = build(default-edge-data: (a:1,b:
[$m_mu$]), {
   node(<a>,id:0)
   edge(source(<a>),id:1)
   edge(sink(<a>),<e>,id:0,b:[#set
text(font:"Reforma")
   This is a test: ])
})
#edges(g).map(e=>e.data.b).join()
```

This is a test: m_μ

Default: `none`

default-source-data `any`

Default data merged into every source half-edge data. Captured source data fields override it.

```
#let g = build(default-source-data: (a:1), {
   node(<a>)
   edge(source(<a>))
   edge(sink(<a>))
})
#edges(g).map(e=>if e.source != none
{e.source.data} else {none})
```

```
((a: 1), none)
```

Default: `none`

default-sink-data `any`

Default data merged into every sink half-edge data. Captured sink data fields override it.

```
#let g = build(default-sink-data: (a:1), {
   node(<a>)
   edge(source(<a>))
   edge(sink(<a>))
})
#edges(g).map(e=>if e.sink != none
{e.sink.data} else {none})
```

```
(none, (a: 1))
```

Default: `none`

parse

Parse one or more DOT graphs into graph objects.

Default data are applied before `eval-*` fields are evaluated, so default data strings can refer to parsed record fields such as `#str(name)`. Parsed fields take precedence over default data fields, so DOT `label="..."` overrides `default-node-data: (label: ...)`.

Linnest uses `dot-parser` for DOT syntax and then gives special meaning to a small set of attributes. Every other attribute is preserved as a flat string statement. Preserved statements are visible through `info()`, `nodes()`, `edges()`, `map()`, and `eval-fields()`, but they do not become native Typst `data` unless a matching `eval-*` argument is passed.

```
#let src = ```dot
digraph { a [label="A"]; }
```
#let n = nodes(parse(src.text).first()).first()
#n.statements.label
#n.data
```

```
A
```

Graph-level handling:

- The DOT graph name becomes `graph.info(g).name`. A graph-level `name` attribute can name an anonymous graph; a named `graph`/`digraph` header takes precedence.

```
#let src = ```dot
digraph header_wins { graph [name="ignored"];
a }
```
#info(parse(src.text).first()).name
```

```
header_wins
```

- Top-level `key=value` statements and `graph [key=value]` attributes become `graph.info(g).global-statements`, except for the consumed `name`. They do not configure layout, even when a key looks like a layout option; pass layout options directly to `layout`.

```
#let src = ```dot
digraph { steps=1; graph [subtitle="nested"];
a }
```
#let statements =
info(parse(src.text).first()).global-statements
#statements.steps
#statements.subtitle
```

```
1 nested
```

- `node [key=value]` and `edge [key=value]` become `default-node-statements` and `default-edge-statements`. They are merged into each parsed node or edge before local attributes, so local attributes override defaults.

```
#let src = ```dot
digraph { node [color=gray]; edge [particle=g];
a [color=red]; a -> b }
```
#let g = parse(src.text).first()
#nodes(g).first().statements.color
#edges(g).first().statements.particle
```

```
red g
```

Node handling:

- The DOT node id becomes the node `name` returned by `nodes()`. If the DOT node id is numeric, it is consumed as the node index instead and the public `name` is `none`.

```
#let src = ```dot
digraph { alpha; 1; }
```
#nodes(parse(src.text).first()).map(n => n.name)
```

```
(<alpha>, none)
```

- `id=<n>` is also consumed as an explicit node index. Explicit node indexes must be unique and in bounds after parsing.

```
#let src = ```dot
digraph { a [id=1]; b [id=0]; }
```
#nodes(parse(src.text).first()).map(n => n.name)
```

```
(<b>, <a>)
```

- `style=invis` marks the DOT node as a dangling external endpoint. It is not returned by `nodes()`. Its remaining attributes are copied onto the external edge that touches it, except `shape` and `label`; `style` and `id` are consumed.

````
#let src = ```dot
digraph { ext [style=invis, column=left,
label=skip]; ext -> a [id=0]; }
```
#let g = parse(src.text).first()
#nodes(g).map(n => n.name)
#edges(g).first().statements.column
#edges(g).first().statements.at("label",
default: none)
````

```
(<a>,) left
```

- Node `style` is consumed for every node; only `style=invis` has special behavior. Use another attribute name for drawing style metadata that must survive parsing.

````
#let src = ```dot
digraph { a [style=filled, "draw-
style"=filled]; }
```
#nodes(parse(src.text).first()).first().statements
````

```
(draw-style: "filled")
```

- `pos` is parsed as node placement. Simple numeric values are exposed as `node.pos`; extended placement values are used by `layout`. `pin` is an explicit layout constraint and is not exposed as a public statement.

````
#let src = ```dot
digraph { a [id=0, pos="0,0!"]; b [id=1,
pos="ref(node:0)+2,0!"]; }
```
#let parsed-nodes =
nodes(parse(src.text).first())
#parsed-nodes.map(n => n.pos)
#parsed-nodes.map(n => n.statements.at("pin",
default: none))
````

```
(none, none) (none, none)
```

- `shift` is parsed as a drawing shift and also remains available as a statement. `eval` is retained for legacy round-tripping. Other node attributes are preserved as node statements.

````
#let src = ```dot
digraph { a [shift="0.1,0", eval=legacy,
label=A]; }
```
#let n = nodes(parse(src.text).first()).first()
#n.shift
#n.statements
````

```
(x: 0.1, y: 0.0) (eval: "legacy", label:
"A", shift: "0.1,0")
```

Edge handling:

- `id=<n>` is consumed as the edge index. It is not preserved as a statement; drawing callbacks expose the resulting stable edge index as `eid`. Explicit edge indexes must be unique and in bounds. Without explicit ids, edge order is parser-internal, not DOT input order.

````
#let src = ```dot
digraph { a -> b [id=1, label=later]; b -> c
[id=0, label=first]; }
```
#let parsed-edges =
edges(parse(src.text).first())
#parsed-edges.map(e => e.edge)
#parsed-edges.map(e => e.statements.label)
#parsed-edges.map(e => e.statements.at("id",
````
````

```
default: none))
```

```
(0, 1) ("first", "later") (none, none)
```

- `dir=forward`, `dir=back`, and `dir=none` become edge orientations `"default"`, `"reversed"`, and `"undirected"`. If omitted, directed DOT edges are `"default"` and undirected DOT edges are `"undirected"`.

```
#let src = ```dot
digraph { a -> b [id=0]; b -> c [id=1,
dir=back]; c -> d [id=2, dir=none]; }
```
#edges(parse(src.text).first()).map(e =>
e.orientation)
```

```
("default", "reversed", "undirected")
```

- `source="..."` and `sink="..."` are consumed as endpoint `statement` values and removed from edge statements. On an external edge, use the attribute matching the real endpoint side in the DOT edge: `ext -> a` uses `sink=...`, while `a -> ext` uses `source=...`.

```
#let src = ```dot
digraph { ext [style=invis]; ext -> a [id=0,
sink=in]; a -> ext [id=1, source=out]; }
```
#let endpoint-statement(endpoint) = if endpoint
== none { none } else { endpoint.at("statement",
default: none) }
#let parsed-edges =
edges(parse(src.text).first())
#parsed-edges.map(e => endpoint-
statement(e.source))
#parsed-edges.map(e => endpoint-
statement(e.sink))
#parsed-edges.map(e => e.statements.at("source",
default: none))
```

```
(none, "out") ("in", none) (none, none)
```

- `pos` is parsed as the edge control-point placement. `pin` is an explicit edge layout constraint. `shift`, `label-pos`, `label-angle`, and `bend` are parsed into edge geometry fields; except for `pin`, they also remain available as statements. Other edge attributes are preserved as edge statements.

```
#let src = ```dot
digraph { a -> b [id=0, pos="0,1!",
pin="x:@edge", shift="0.1,0", "label-
pos"="0,1.2", "label-angle"="0.3rad",
bend="0.4rad", particle=g]; }
```
#let e = edges(parse(src.text).first()).first()
#e.pos
#e.shift
#e.label-pos
#e.label-angle
#e.bend
#e.statements.particle
#e.statements.at("pin", default: none)
```

```
(x: 0.1, y: 0.0) (x: 0.0, y: 1.2) 0.3
0.4 g
```

- Attributes copied from an invisible endpoint node are merged into the external edge statements unless their keys are `shape` or `label`.
```

```
#let src = ```dot
digraph { ext [style=invis, column=left,
shape=none, label=skip]; ext -> a [id=0]; }
```
#edges(parse(src.text).first()).first().statements
```

```
(column: "left")
```

Port and half-edge handling:

- DOT ports of the form `node:port` and `node:port:compass` are preserved on source/sink endpoint records. Numeric ports also assign explicit half-edge ids. Non-numeric ports are exposed only as `port-label`.

```
#let src = ```dot
digraph { a:left:e -> b:0:w [id=0]; }
```
#let e = edges(parse(src.text).first()).first()
#e.source.port-label
#e.sink.hedge
```

```
left 0
```

- Compass values are exposed as `compass`; valid DOT compass names include `n`, `ne`, `e`, `se`, `s`, `sw`, `w`, `nw`, `c`, and `_`.

```
#let src = ```dot
digraph { a:n -> b:sw [id=0]; }
```
#let e = edges(parse(src.text).first()).first()
#e.source.compass
#e.sink.compass
```

```
n s
```

- Explicit half-edge ids from numeric ports must be unique and in bounds. They are retained for half-edge ordering and for `join()` with `key: "id"`; query and eval records expose the resulting half-edge index as `hedge`.

```
#let src = ```dot
digraph { a:1 -> b:0 [id=0]; }
```
#let e = edges(parse(src.text).first()).first()
#e.source.hedge
#e.sink.hedge
```

```
1 0
```

Placement fields:

- `pos="x,y"` gives a starting coordinate; `pos="x,y!"` pins both axes.

```
#let src = ```dot
digraph { a [id=0, pos="0,0"]; b [id=1,
pos="2,0!"]; }
```
#let parsed-nodes =
nodes(parse(src.text).first())
#parsed-nodes.map(n => n.pos)
#parsed-nodes.map(n => n.statements.at("pos-
mode", default: none))
```

```
((x: 0.0, y: 0.0), none) (none, none)
```

- `pos="ref(node:<id>)+dx,dy!"` and `pos="ref(edge:<id>)+dx,dy!"` reference explicit DOT node or edge ids, not names and not implicit parse order.

```
#let src = ```dot
digraph { a [id=0, pos="1,1!"]; b [id=1,
pos="ref(node:0)+2,0!"]; a -> b [id=0,
pos="ref(node:1)+0,1!"]; }
```
#let g = parse(src.text).first()
#nodes(g).at(1).pos
```

```
#edges(g).first().pos
```

- Axis form accepts `x:<coord>` and `y:<coord>` entries. `!` applies to the individual axis, for example `pos="x:2!,y:0"`.

```
#let src = ```dot
digraph { a [id=0, pos="x:2!,y:0"]; }
```
#nodes(parse(src.text).first()).first().pos
```

- Group coordinates use `@name`, `@+name`, or `@-name` in axis form and must be pinned with `!`, for example `pos="x:@-left!,y:@row!"`.

```
#let src = ```dot
digraph { ext [style=invis]; ext -> a [id=0,
pos="x:@-left!,y:@row!"]; }
```
#edges(layout(parse(src.text).first(), layout-
algo: "tree")).first().pos
```

```
(x: -1.575, y: 1.0)
```

- `pin` accepts numeric point constraints, `x:<coord>`, `y:<coord>`, and grouped constraints such as `x:@left`, `x:@+right`, or `@row`.

```
#let src = ```dot
digraph { a -> b [id=0, pin="x:@+right"]; }
```
#edges(layout(parse(src.text).first(), layout-
algo: "tree")).first().statements.at("pin",
default: none)
```

**Parameters**

```
parse(
  input: string bytes,
  default-node-data: any,
  eval-node-fields: string array,
  default-edge-data: any,
  eval-edge-fields: string array,
  default-source-data: any,
  eval-source-fields: string array,
  default-sink-data: any,
  eval-sink-fields: string array,
  eval-graph-fields: string array,
  eval-mode: string,
  scope: dictionary
) -> array
```

input `string` or `bytes`

DOT source text containing one or more `graph` or `digraph` definitions.
```

default-node-data `any`

Default data merged into every node data. Captured node data fields override it, but bare dot statements don't set data fields. To turn statements into data fields, use `eval-node-fields`.

```
#let a = ```dot
digraph {
a -> b -> c -> d -> a
a -> a
b -> d
c [particle="q"]
}
```
#let g = parse(a.text,default-node-data:
(particle:"g")).at(0)
#nodes(g).map(n=>n.data.particle)
```

```
("g", "g", "g", "g")
```

Default: `none`

eval-node-fields `string` or `array`

Node fields to evaluate into node data. The eval scope includes node statements plus `node`, `name`, and `pos`.

```
#let src = ```dot
digraph { a [label="#str(name)"]; }
```
#nodes(parse(src.text, eval-node-fields:
"label").first()).first().data.label
```

```
a
```

Default: `()`

default-edge-data `any`

Default data merged into every edge data. Captured edge data fields override it, but bare dot statements don't set data fields. To turn statements into data fields, use `eval-edge-fields`.

Default: `none`

eval-edge-fields `string` or `array`

Edge statement fields to evaluate into `graph.edges(g).at(i).data`. The eval scope includes edge statements plus `edge`, `orientation`, endpoint records, placement fields, and existing edge data. Drawing callbacks later expose the edge index as `eid`.

```
#let src = ```dot
digraph { a -> b [id=0,
label="#orientation"]; }
```
#edges(parse(src.text, eval-edge-fields:
"label").first()).first().data.label
```

```
default
```

Default: `()`
```

default-source-data `any`

Default data merged into every source half-edge data. Captured source data fields override it.

Default: `none`

eval-source-fields `string` or `array`

Source half-edge fields to evaluate into `edge.source.data`. The eval scope includes source endpoint fields `statement`, `port-label`, `compass`, `node`, and `hedge`, plus the surrounding edge fields.

```
#let src = ```dot
digraph { a:left:e -> b [id=0, source="#port-
label"]; }
```
#edges(parse(src.text, eval-source-fields:
"statement").first()).first().source.data.statem
```

```
left
```

Default: `()`

default-sink-data `any`

Default data merged into every sink half-edge data. Captured sink data fields override it.

Default: `none`

eval-sink-fields `string` or `array`

Sink half-edge fields to evaluate into `edge.sink.data`. The eval scope includes sink endpoint fields `statement`, `port-label`, `compass`, `node`, and `hedge`, plus the surrounding edge fields.

```
#let src = ```dot
digraph { a -> b:0:w [id=0,
sink="#str(hedge)"]; }
```
#edges(parse(src.text, eval-sink-fields:
"statement").first()).first().sink.data.statemen
```

```
0
```

Default: `()`

eval-graph-fields `string` or `array`

Graph statement fields to evaluate into `graph.info(g).data`. The eval scope includes graph statements and direct graph record fields.

```
#let src = ```dot
digraph {
  title = "Strong graph";
}
```
#info(parse(src.text, eval-graph-fields:
"title").first()).data.title
```

```
Strong graph
```

Default: `()`
```

eval-mode `string`

Typst `eval` mode used for string field values.

Default: `"markup"`

scope `dictionary`

Additional Typst names available while evaluating field values.

Default: `(:)`

node

Create a graph node item for `build()`.

A Typst label is the node name used by `source()`, `sink()`, and `pos()`. The optional numeric `id` fixes the resulting graph node index. Extra named arguments are captured as node data fields, so `node(<a>, label: [A], color: red)` stores `(label: [A], color: red)`. The default draw style uses `data.label` as the visible node label when present.

Parameters

```
node(
  ..args: label,
  name: none label,
  id: none int,
  pos: none dictionary,
  shift: none string array dictionary,
  statements: dictionary
) -> array
```

..args `label`

Optional positional node name. Must be a Typst label when provided; extra named arguments become data fields.

name `none` or `label`

Typst node name for references.

Default: `none`

id `none` or `int`

Numeric graph node index. Must be unique and in bounds when provided.

Default: `none`

pos `none` or `dictionary`

Node placement.

Default: `none`

shift `none` or `string` or `array` or `dictionary`

Drawing shift stored as a statement.

Default: `none`

statements `dictionary`

Additional flat node statements. Used by DOT; values cannot nest.

Default: `(:)`

source

Create a source half-edge endpoint.

`node` may be a node name like <a> or a numeric node index. `name` gives the half-edge a Typst name; `id` is a numeric half-edge order/index override. Extra named arguments are captured as source data fields.

Parameters

```
source(
  node: label int,
  ..args: any,
  name: none label,
  id: none int,
  statement: none string,
  compass: none string
) -> dictionary
```

node `label` or `int`

Referenced node, either by Typst label name or numeric node index.

..args `any`

Extra named arguments become source data fields.

name `none` or `label`

Optional half-edge name used for later references.

Default: `none`

id `none` or `int`

Optional numeric half-edge index/order override.

Default: `none`

statement `none` or `string`

DOT-ish flat statement used for matching/joining dangling half edges.

Default: `none`

compass `none` or `string`

DOT compass point such as `"n"`, `"s"`, `"e"`, or `"w"`.

Default: `none`

sink

Create a sink half-edge endpoint.

`node` may be a node name like `<a>` or a numeric node index. `name` gives the half-edge a Typst name; `id` is a numeric half-edge order/index override. Extra named arguments are captured as sink data fields.

Parameters

```
sink(
  node: label int,
  ..args: any,
  name: none label,
  id: none int,
  statement: none string,
  compass: none string
) -> dictionary
```

node `label` or `int`

Referenced node, either by Typst label name or numeric node index.

..args `any`

Extra named arguments become sink data fields.

name `none` or `label`

Optional half-edge name used for later references.

Default: `none`

id `none` or `int`

Optional numeric half-edge index/order override.

Default: `none`

statement `none` or `string`

DOT-ish flat statement used for matching/joining dangling half edges.

Default: `none`

compass `none` or `string`

DOT compass point such as `"n"`, `"s"`, `"e"`, or `"w"`.

Default: `none`

edge

Create a graph edge item for `build()`.

Positional arguments may contain one `source()`, one `sink()`, and optionally one Typst label used as the edge name. The numeric `id` chooses the edge order. Extra named arguments are captured as edge data fields, so `edge(source(<a>), sink(<b>), particle: "g")` stores `(particle: "g")`. The default draw style uses `data.label` as the visible edge label when present.

Parameters

```
edge(
  ..args: any,
  name: none label,
  id: none int,
  orientation: string,
  pos: none dictionary,
  shift: none string array dictionary,
  label-pos: none string array dictionary,
  label-angle: none int float string,
  bend: none int float string,
  statements: dictionary
) -> array
```

..args `any`

Source/sink half-edges and optional edge name; extra named arguments become data fields.

name `none` or `label`

Typst edge name.

Default: `none`

id `none` or `int`

Numeric edge order/index override.

Default: `none`

orientation `string`

Edge orientation: `"default"`, `"reversed"`, or `"undirected"`.

Default: `"default"`

pos `none` or `dictionary`

Edge placement.

Default: `none`

shift `none` or `string` or `array` or `dictionary`

Drawing shift stored as a statement.

Default: `none`

label-pos `none` or `string` or `array` or `dictionary`

Edge label position stored as a statement.

Default: `none`

label-angle `none` or `int` or `float` or `string`

Edge label angle stored as a statement.

Default: `none`

bend `none` or `int` or `float` or `string`

Edge bend stored as a statement.

Default: `none`

statements `dictionary`

Additional flat edge statements. Used by DOT; values cannot nest.

Default: `(:)`

group

Create a grouped placement coordinate.

`side: "+"` keeps the solved coordinate non-negative and `side: "-"` keeps it non-positive. Groups are layout constraints and therefore require pin placement, which is the `pos()` default.

```
#group("right", side: "+")
```

```
(kind: "group", name: "right", side: "+")
```

Parameters

```
group(
  name: string  int  bool,
  side: none  string
) -> dictionary
```

name `string` or `int` or `bool`

Group identifier shared by positions constrained to the same coordinate.

side `none` or `string`

Optional sign constraint: `"+"`, `"-"`, `"positive"`, or `"negative"`.

Default: `none`

pos

Create a first-class graph placement.

The default `mode: "pin"` turns numeric and grouped coordinates into layout constraints and also makes the coordinates immediately drawable without a layout pass. Use `mode: "start"` when the coordinate should only seed the layout.

```
#pos(x: 0, y: group("row"), mode: "pin")
```

```
(
  mode: "pin",
  x: 0,
  y: (kind: "group", name: "row", side:
none),
)
```

Parameters

```
pos(
  x: none int float dictionary,
  y: none int float dictionary,
  ref: none label int,
  dx: none int float,
  dy: none int float,
  mode: string
) -> dictionary
```

x `none` or `int` or `float` or `dictionary`

Absolute or grouped x coordinate.

Default: `none`

y `none` or `int` or `float` or `dictionary`

Absolute or grouped y coordinate.

Default: `none`

ref `none` or `label` or `int`

Node reference for relative placement, by name or numeric index.

Default: `none`

dx `none` or `int` or `float`

Relative x offset from `ref`.

Default: `none`

dy `none` or `int` or `float`

Relative y offset from `ref`.

Default: `none`

mode `string`

Placement mode: `"pin"` constrains layout, `"start"` only seeds it.

Default: `"pin"`

map

Map graph metadata to new native data.

The callbacks receive decoded records plus a `fields` dictionary containing merged statements and direct record fields. A callback returns `none` to leave the record unchanged, or `(data: value)` to set new native data.

```
#let g = build({ node(<a>) })
#let g = map(g, node: node => (data: (label:
[A])))
#nodes(g).first().data.label
```

```
A
```

Parameters

```
map(
  graph_: dictionary,
  graph: none function,
  node: none function,
  edge: none function,
  source: none function,
  sink: none function
) -> dictionary
```

graph_ `dictionary`

Graph object to transform.

graph `none` or `function`

Callback for graph metadata records.

Default: `none`

node `none` or `function`

Callback for node records.

Default: `none`

edge `none` or `function`

Callback for edge records.

Default: `none`

source `none` or `function`

Callback for source half-edge records.

Default: `none`

sink `none` or `function`

Callback for sink half-edge records.

Default: `none`

eval-fields

Evaluate selected fields into native data entries.

Each selected field is read from the record's merged `fields` dictionary, evaluated in a scope containing those fields, and written to `data.<field>`.

```
#let g = build({
  node(<a>)
  node(<b>)
  edge(source(<a>), sink(<b>), statements: (mom:
"p"))
}, default-edge-statements: (display-label:
"$#mom$"))
#let g = eval-fields(g, eval-edge-fields:
("display-label",))
#edges(g).first().data.at("display-label")
```

p

Parameters

```
eval-fields(
  graph_: dictionary,
  eval-graph-fields: string array,
  eval-node-fields: string array,
  eval-edge-fields: string array,
  eval-source-fields: string array,
  eval-sink-fields: string array,
  eval-mode: string,
  scope: dictionary
) -> dictionary
```

graph_ `dictionary`

Graph object whose selected statement fields should be evaluated.

eval-graph-fields `string` or `array`

Graph statement fields to evaluate into `graph.info(g).data`.

Default: `()`

eval-node-fields `string` or `array`

Node statement fields to evaluate into node data.

Default: `()`

eval-edge-fields `string` or `array`

Edge statement fields to evaluate into edge data.

Default: `()`

eval-source-fields `string` or `array`

Source half-edge fields to evaluate into source data.

Default: `()`

eval-sink-fields `string` or `array`

Sink half-edge fields to evaluate into sink data.

Default: `()`

eval-mode `string`

Typst `eval` mode used for string field values.

Default: `"markup"`

scope `dictionary`

Additional Typst names available while evaluating field values.

Default: `(:)`

info

Return graph metadata.

The result has `name`, `global-statements`, `default-edge-statements`, and `default-node-statements`.

```
#let g = build({ node(<a>) }, name: "demo")
#info(g).name
```

```
demo
```

Parameters

`info(graph: dictionary) -> dictionary`

graph `dictionary`

Graph object returned by `build()`, `parse()`, `layout`, or another graph API.

dot

Serialize a graph object to DOT.

```
#let g = build({
  node(<a>)
  node(<b>)
  edge(source(<a>), sink(<b>))
}, name: "demo")
#dot(g).contains("digraph demo")
```

```
true
```

Parameters

`dot(graph: dictionary) -> string`

graph `dictionary`

Graph object to serialize.

nodes

Return node records, optionally filtered by a subgraph object.

Node `name` values are Typst labels when present.

```
#let g = build({
  node(<a>)
  node(<b>)
})
#nodes(g).map(node => str(node.name)).join(", ")
```

```
a, b
```

Parameters

```
nodes(
  graph: dictionary,
  subgraph: none bytes
) -> array
```

graph `dictionary`

Graph object to inspect.

subgraph `none` or `bytes`

Optional subgraph filter; only nodes incident to selected half edges are returned.

Default: `none`

edges

Return edge records, optionally filtered by a subgraph object.

Edge `name` values are Typst labels when present.

```
#let g = build({
  node(<a>)
  node(<b>)
  edge(source(<a>, compass: "e"), sink(<b>))
})
#edges(g).len()
```

```
1
```

Parameters

```
edges(
  graph: dictionary,
  subgraph: none bytes
) -> array
```

graph `dictionary`

Graph object to inspect.

subgraph `none` or `bytes`

Optional subgraph filter; only selected edges/half-edges are returned.

Default: `none`

node-data

Return one named node's native data.

`name` is a Typst label such as <a> or the corresponding string name.

```
#let g = build({ node(<a>, label: [A]) })
#node-data(g, <a>).label
```

A

Parameters

```
node-data(
  graph_: dictionary,
  name: label string
) -> any
```

graph_ `dictionary`

Graph object to inspect.

name `label` or `string`

Node name as a Typst label or its string form.

edge-data

Return one named edge's native data.

`name` is a Typst label such as <e> or the corresponding string name.

```
#let g = build({
  node(<a>)
  node(<b>)
  edge(source(<a>), <e>, sink(<b>), label:
[$p$])
})
#edge-data(g, <e>).label
```

p

Parameters

```
edge-data(
  graph_: dictionary,
  name: label string
) -> any
```

graph_ `dictionary`

Graph object to inspect.

name `label` or `string`

Edge name as a Typst label or its string form.

update-node-data

Update one named node's native data.

`update` may be a replacement data value or a function `(data, node) => new-data`.

```
#let g = build({ node(<a>) })
#let g = update-node-data(g, <a>, (label: [A]))
#nodes(g).first().data.label
```

A

Parameters

```
update-node-data(
  graph_: dictionary,
  name: label string,
  update: any function
) -> dictionary
```

graph_ `dictionary`

Graph object to update.

name `label` or `string`

Node name as a Typst label or its string form.

update `any` or `function`

Replacement data or `(data, node) => new-data` callback.

update-edge-data

Update one named edge's native data.

`update` may be a replacement data value or a function `(data, edge) => new-data`.

```
#let g = build({
  node(<a>)
  node(<b>)
  edge(source(<a>), <e>, sink(<b>))
})
#let g = update-edge-data(g, <e>, (label:
[$p$]))
#edges(g).first().data.label
```

p

Parameters

```
update-edge-data(
  graph_: dictionary,
  name: label string,
  update: any function
) -> dictionary
```

graph_ `dictionary`

Graph object to update.

name `label` or `string`

Edge name as a Typst label or its string form.

update `any` or `function`

Replacement data or `(data, edge) => new-data` callback.

join

Join two graphs by matching dangling half-edge statements or ids on `key`.

Supported key values are `"statement"`, `"compass"`, and `"id"`.

```
#let left = build({
  node(<a>)
  edge(sink(<a>, statement: "j"))
})
#let right = build({
  node(<b>)
  edge(source(<b>, statement: "j"))
})
#edges(join(left, right, key:
"statement")).len()
```

```
1
```

Parameters

```
join(
  left: dictionary,
  right: dictionary,
  key: string
) -> dictionary
```

left `dictionary`

Left graph object.

right `dictionary`

Right graph object.

key `string`

Dangling half-edge match key: `"statement"`, `"compass"`, or `"id"`.

Default: `"statement"`

cycles

Return subgraph objects for the graph's cycle basis.

```
#let g = build({
  node(<a>)
  node(<b>)
  edge(source(<a>), sink(<b>))
})
#cycles(g).len()
```

```
0
```

Parameters

```
cycles(graph: dictionary) -> array
```

graph `dictionary`

Graph object to analyze.

forests

Return subgraph objects for the graph's spanning forests.

```
#let g = build({
  node(<a>)
  node(<b>)
  edge(source(<a>), sink(<b>))
})
#forests(g).len()
```

```
1
```

Parameters

`forests(graph: dictionary) -> array`

graph `dictionary`

Graph object to analyze.

layout

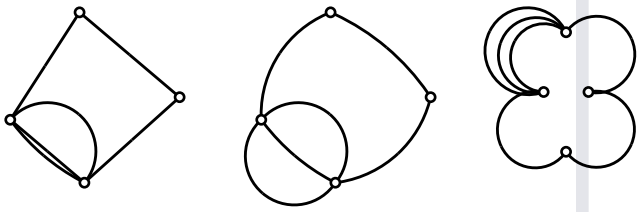
- layout()

layout

Apply the linnest layout pass to a graph object.

This is intentionally a second step: construct or parse a graph first, then call `layout`. Set `layout-algo` to `"force"` for deterministic force integration, `"anneal"` for simulated annealing, `"tree"` for a traversal tree placement, or `"dot"` for a layered directed placement.

```
#let g = graph.parse("digraph partial { a -> b;a
-> b;a:s -> b:s; b:s -> c:s; c:s -> d:s; d:s ->
a:s }").at(0)
#let south = subgraph.compass(g,"s")
#let gf = layout(layout(g, layout-algo:
"force"), layout-algo: "tree", layout-nodes:
"fixed",subgraph:south)
#let ga = layout(g, layout-algo: "force")
#let gt = layout(layout(g, layout-algo:
"tree",layout-roots: (2)), layout-algo:
"anneal", layout-nodes: "fixed",gamma-
ee:0.1,gamma-ev:.75,beta:5,length-scale:0.4)
#grid(columns: 3, gutter: 2cm, draw(gf),
draw(ga), draw(gt))
```



```
#let g = graph.parse("digraph partial { a -> b;
b -> c; c -> d; d -> a }").at(0)
#let tree = graph.forests(g).at(0)
#let g = layout(g, layout-algo: "tree",
subgraph: tree)
#graph.edges(g).map(edge => edge.pos)
```

```
(
  (x: 0.0, y: 1.05),
  (x: 0.0, y: 3.1500000000000004),
  (x: 0.0, y: 5.25),
  (x: 0.0, y: 3.1500000000000004),
)
```

Parameters

```
layout(
  graph: dictionary,
  subgraph: none bytes,
  viewport-w: float,
  viewport-h: float,
  tree-dx: float,
  tree-dy: float,
  steps: int,
  seed: int,
  step: float,
  step-shrink: float,
  cool: float,
  accept-floor: float,
  early-tol: float,
  temp: float,
  delta: float,
  beta: float,
  k-spring: float,
  g-center: float,
  epochs: int,
  crossing-penalty: float,
  gamma-dangling: float,
  gamma-ee: float,
  directional-force: float,
  label-length-scale: float,
  label-spring: float,
  label-charge: float,
  label-steps: int,
  label-layout: string,
  label-step: float,
  label-early-tol: float,
  label-max-delta-scale: float,
  gamma-ev: float,
  eps: float,
  incremental-energy: bool,
  layout-algo: string,
  layout-nodes: string,
  layout-roots: array,
  z-spring: float,
  z-spring-growth: float,
  length-scale: float
) -> dictionary
```

graph `dictionary`

Graph object returned by `graph.build` or `graph.parse`.

subgraph `none` or `bytes`

Optional subgraph object to lay out. With `"tree"` and `"dot"`, other edges are drawn from the resulting node positions as straight lines. With `"force"` and `"anneal"`, nodes and edges outside the subgraph are fixed boundary points during optimization.

Default: `none`

viewport-w `float`

Width of the layout viewport used to derive the natural spring length. Applies to both `"force"` and `"anneal"`.

Default: `10.0`

viewport-h `float`

Height of the layout viewport used to derive the natural spring length. Applies to both `"force"` and `"anneal"`.

Default: `10.0`

tree-dx `float`

Horizontal spacing multiplier for the initial traversal-tree placement. Applies before both `"force"` and `"anneal"`.

Default: `0.9`

tree-dy `float`

Vertical spacing multiplier for the initial traversal-tree placement. Applies before both `"force"` and `"anneal"`.

Default: `1.2`

steps `int`

Iterations per epoch. In `"force"` mode this is the number of force integration steps; in `"anneal"` mode this is the number of proposals per temperature epoch.

Default: `int(sys.inputs.at("steps", default: "30"))`

seed `int`

Seed for deterministic initialization, force-mode jitter, and annealing proposals. Applies to both modes.

Default: `int(sys.inputs.at("seed", default: "2"))`

step `float`

Initial movement scale. `"force"` multiplies computed forces by this value; `"anneal"` uses it as the proposal step size.

Default: `0.81`

step-shrink `float`

Anneal-only step shrink factor, applied when an epoch's acceptance ratio falls below `accept-floor`.

Default: `0.21`

cool `float`

Cooling factor applied once per epoch. `"anneal"` cools `temp`; `"force"` shrinks `step`.

Default: `0.85`

accept-floor `float`

Anneal-only acceptance-ratio threshold below which `step` is shrunk by `step-shrink`.

Default: `0.15`

early-tol `float`

Force-only early stop threshold for maximum movement in one step. The annealing schedule stores this value but does not currently use it for stopping.

Default: `1e-6`

temp `float`

Anneal-only initial temperature used in the Metropolis acceptance test.

Default: `0.3`

delta `float`

Force-mode maximum movement clamp per point and per step.

Default: `0.4`

beta `float`

Base repulsion strength for vertex-vertex interactions. Also scales `gamma-ev`, `gamma-ee`, `gamma-dangling`, and `g-center`. Applies to both modes through the shared spring energy.

Default: `50.0`

k-spring `float`

Spring stiffness for node-to-edge incidence lengths. Applies to both modes.

Default: `11.0`

g-center `float`

Centering strength relative to `beta`. Applies to both modes.

Default: `0.002`

epochs `int`

Number of epochs. Both modes run up to `steps` iterations inside each epoch.

Default: `30`

crossing-penalty `float`

Anneal-only fixed energy penalty per detected edge crossing. The direct force integrator does not currently add a crossing force.

Default: `30.0`

gamma-dangling `float`

Repulsion for dangling half edges, relative to `beta`. Applies to both modes through the shared spring energy.

Default: `5.0`

gamma-ee `float`

Local edge-edge repulsion, relative to `beta`. Applies to both modes.

Default: `0.1`

directional-force `float`

Bias that pushes points in directions implied by pin/port constraints. Applies to both modes.

Default: `5.0`

label-length-scale `float`

Edge-label target offset as a multiple of the graph spring length. Label layout runs after both graph layout modes.

Default: `0.6`

label-spring `float`

Spring strength pulling each label toward its target offset in the spring-based label layouts.

Default: `23.0`

label-charge `float`

Repulsion strength between labels and graph points, scaled by spring length squared. Label layout runs after both modes.

Default: `3.0`

label-steps `int`

Maximum number of post-layout label relaxation steps. Set to `0` to skip label placement. Applies after both modes.

Default: `20`

label-layout `string`

Edge-label relaxation model. `"normal"` uses a perpendicular offset, `"dangling-tangent"` uses the edge direction for dangling half-edge labels and a perpendicular offset for paired edges, and `"fixed-length"` keeps each label at a fixed distance from its edge point and only lets that segment rotate.

Default: `"normal"`

label-step `float`

Label relaxation step size. Applies after both modes.

Default: `0.15`

label-early-tol `float`

Label relaxation early stop threshold. Applies after both modes.

Default: `1e-3`

label-max-delta-scale `float`

Label movement clamp as a multiple of spring length. Applies after both modes.

Default: `0.5`

gamma-ev `float`

Edge-vertex repulsion, relative to `beta`. Applies to both modes.

Default: `0.01`

eps `float`

Softening epsilon used in inverse-square force/energy terms. Applies to both modes.

Default: `1e-4`

incremental-energy `bool`

Whether annealing updates cached energy by local deltas. This is anneal-only; force mode computes direct forces instead of energies.

Default: `true`

layout-algo `string`

Layout algorithm. Use `"force"` for direct force integration, `"anneal"` for simulated annealing against the spring energy, `"tree"` for a traversal-tree placement, or `"dot"` for a layered directed placement.

Default: `"force"`

layout-nodes `string`

Node movement policy. `"layout"` lets the layout algorithm move nodes. `"fixed"` keeps every node at its current position for this layout pass and only moves edge control points. With `subgraph`, only edges in the subgraph are moved; other edge control points stay at their current positions.

Default: `"layout"`

layout-roots `array`

Ordered node indices used as preferred roots for `"tree"` and `"dot"`. Roots outside the selected node set are ignored. Remaining components are laid out afterward in graph order.

Default: `()`

z-spring `float`

Force-only spring pulling temporary z coordinates back toward the layout plane. Use with `z-spring-growth` to help separate overlapping points during integration.

Default: `0.05`

z-spring-growth `float`

Force-only per-epoch multiplier for `z-spring`.

Default: `1.3`

length-scale `float`

Natural spring-length multiplier. This scales the graph's preferred edge length and the repulsive coefficients derived from it. Applies to both modes.

Default: `0.35`

physics

- stroke-style()
- source-stroke()
- sink-stroke()
- particle-name()
- edge-entry()
- text-value()
- eval-expression()
- label-content()
- style-dict()
- momentum-value()
- edge-index()
- dangling-half-edge-index()
- source-style()
- sink-style()
- edge-label()
- style()

Variables

- massive
- massless

- dashed
- dotted
- fermion-arrow-mark
- fermion-flow
- wave
- coil
- zigzag
- default-edge
- default-map
- momentum-arrow-defaults

stroke-style

Build a rounded CeTZ stroke dictionary accepted by `linnest.draw`.

Parameters

```
stroke-style(
  c,
  thickness,
  dash
) -> dictionary
```

source-stroke

Source-half stroke helper.

Parameters

```
source-stroke(
  c,
  thickness,
  dash
) -> dictionary
```

sink-stroke

Sink-half stroke helper. The default lightening makes the two halves encode the graph's source/sink split without requiring arrowheads.

Parameters

```
sink-stroke(
  c,
  thickness,
  dash,
  lighten
) -> dictionary
```

particle-name

Return an edge's particle name, stripping the quotes often present in DOT statement values.

Parameters

```
particle-name(edge) -> none string
```

edge-entry

Return the style-map entry for an edge.

Parameters

```
edge-entry(
  edge,
  map,
  default
) -> dictionary
```

text-value

Convert DOT-ish values to plain text.

Parameters

```
text-value(value) -> string
```

eval-expression

Evaluate a string-valued style expression in the edge scope.

Parameters

```
eval-expression(
  value,
  edge,
  map,
  scope
) -> any
```

label-content

Convert a style label value to content. In `mode: "eval"`, string values are evaluated in the edge scope.

Parameters

```
label-content(
  value,
  edge,
  mode,
  map,
  scope
) -> none content
```

style-dict

Convert a style value to a dictionary. In `mode: "eval"`, string values are evaluated in the edge scope.

Parameters

```
style-dict(
  value,
  edge,
  mode,
  map,
  scope
) -> dictionary
```

momentum-value
Return the momentum field used by optional edge labels.

Parameters
```
momentum-value(
  edge,
  fields
) -> none any
```

edge-index
Return the edge index used by optional edge labels. The DOT `id` statement wins over the renderer-local `eid`.

Parameters
```
edge-index(
  edge,
  fields
) -> none any
```

dangling-half-edge-index
Return the exposed half-edge index for dangling edges only.

Parameters
```
dangling-half-edge-index(edge) -> none any
```

source-style
Style callback for the source half edge.

`momentum-arrows: true` adds one centered black CeTZ-mark decoration while the main edge remains drawn with its normal particle style.

Parameters
```
source-style(
  edge,
  map,
  default,
  typst-fields,
  scope,
  orientation-split,
  momentum-arrows,
  momentum-arrow-offset,
  momentum-arrow-length,
  momentum-arrow-ratio,
  momentum-arrow-stroke,
  momentum-arrow-mark
) -> dictionary array
```

sink-style
Style callback for the sink half edge.

`momentum-arrows: true` adds one centered black CeTZ-mark decoration toward the sink node, so momentum arrows always flow from source to sink independently of `edge.orientation`.

Parameters

```
sink-style(
  edge,
  map,
  default,
  typst-fields,
  scope,
  orientation-split,
  momentum-arrows,
  momentum-arrow-offset,
  momentum-arrow-length,
  momentum-arrow-ratio,
  momentum-arrow-stroke,
  momentum-arrow-mark
) -> dictionary array
```

edge-label

Edge-label callback.

By default this preserves data `display-label` or `label`, then explicit `display-label`, `label`, and particle-map labels. Set any `show-*` option to build a label from selected metadata instead:

```
#let callbacks = physics.style(
  momentum-arrows: true,
  show-edge-index: true,
  show-particle: true,
)
```

Parameters

```
edge-label(
  edge,
  map,
  default,
  typst-fields,
  scope,
  show-momentum,
  show-edge-index,
  show-half-edge-index,
  show-particle,
  momentum-fields,
  edge-index-fields,
  momentum-prefix,
  edge-index-prefix,
  half-edge-index-prefix,
  particle-prefix,
  label-separator,
  ],
  label-size,
  label-fill
) -> none content
```

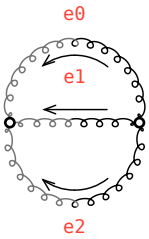
style

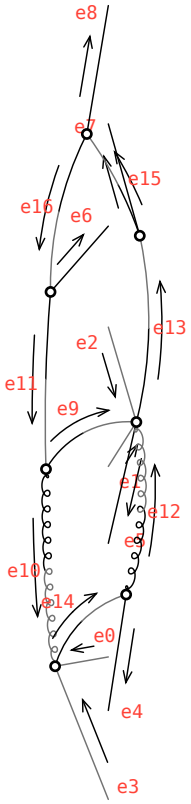
Bundle source-style, sink-style, and edge-label callbacks with shared options for `linnest.draw`.

```
 #let g = build({
   node(<a>,pos:pos(y:0))
   node(<b>,pos:pos(y:0))
   edge(source(<a>), <g-edge>, sink(<b>),
particle: "g")
   edge(source(<a>), <g-edge2>, sink(<b>),
particle: "g")
   edge(source(<a>), <g-edge3>, sink(<b>),
particle: "g")
 },
   name: "physics",
 )
 #let a = ```dot
 digraph {
 0 [dod="-100" num="1"];
  1 [dod="-100" num="1"];
  2 [dod="-100" num="1"];
  3 [dod="-100" num="1"];
  4 [dod="-100" num="1"];
  5 [dod="-100" num="1"];
  6 [dod="-100" num="1"];
  exte0  [style=invis];
  exte0 -> 2:0   [id=0  dod="-100"
lmb_rep="P(0,a___)" num="1" particle="d"
pin="x:@-left"];
  exte1  [style=invis];
  exte1 -> 1:1   [id=1 dir=back  dod="-100"
lmb_rep="P(1,a___)" num="1" particle="d~"
pin="x:@-left"];
  exte2  [style=invis];
  exte2 -> 1:2   [id=2 dir=none  dod="-100"
lmb_rep="P(2,a___)" mass="0" num="1"
particle="H" pin="x:@-left"];
  exte3  [style=invis];
  exte3 -> 2:3   [id=3 dir=none  dod="-100"
is_dummy="true" lmb_rep="0" mass="0" num="1"
particle="H" pin="x:@-left"];
  exte4  [style=invis];
  6:4 -> exte4   [id=4 dir=none  dod="-100"
lmb_rep="P(3,a___)" mass="0" num="1"
particle="H" pin="x:@+right"];
  exte5  [style=invis];
  1:5 -> exte5   [id=5 dir=none  dod="-100"
is_dummy="true" lmb_rep="0" mass="0" num="1"
particle="H" pin="x:@+right"];
  exte6  [style=invis];
  5:6 -> exte6   [id=6 dir=none  dod="-100"
lmb_rep="P(4,a___)" num="1" particle="H"
pin="x:@+right"];
  exte7  [style=invis];
  4:7 -> exte7   [id=7 dir=none  dod="-100"
lmb_rep="P(5,a___)" num="1" particle="H"
pin="x:@+right"];
  exte8  [style=invis];
  3:8 -> exte8   [id=8 dir=none  dod="-100"
lmb_rep="-1*P(3,a___)+-1*P(4,a___)+-1*P(5,a___)+P(0,a___)+P(1,a___)+P(2,a___)"
num="1" particle="H" pin="x:@+right"];
  0:9 -> 1:10  [id=9  dod="-100" lmb_id="1"
lmb_rep="K(1,a___)" num="1" particle="t"];
  0:11  -> 2:12  [id=10 dir=none  dod="-100"
lmb_rep="-1*P(0,a___)+K(0,a___)" num="1"
particle="g"];
```

```
  5:13  -> 0:14  [id=11  dod="-100"
lmb_rep="-1*P(0,a___)+K(0,a___)+K(1,a___)"
num="1" particle="t"];
  6:24  -> 1:18  [id=12 dir=none  dod="-100"
lmb_rep="-1*P(3,a___)+K(0,a___)"
mass="((Q(2,spenso::cind(0)))^2+(Q(2,spenso::cind(1)))^2*-1+(Q(2,spenso::cind(2)))^2*-1+(Q(2,spenso::cind(3)))^2*-
num="1" particle="g"];
  1:15  -> 4:16  [id=13  dod="-100"
lmb_rep="-1*P(3,a___)+K(0,a___)+K(1,a___)+P(1,a___)+P(2,a___)"
num="1" particle="t"];
  2:17  -> 6:23  [id=14  dod="-100" lmb_id="0"
lmb_rep="K(0,a___)" num="1" particle="d"];
  4:19  -> 3:20  [id=15  dod="-100"
lmb_rep="-1*P(3,a___)+-1*P(5,a___)+K(0,a___)+K(1,a___)+P(1,a___)+P(2,a___)"
num="1" particle="t"];
  3:21  -> 5:22  [id=16  dod="-100"
lmb_rep="-1*P(0,a___)+K(0,a___)+K(1,a___)+P(4,a___)"
num="1" particle="t"];
}

 ```
 #let (source-style, sink-style, edge-
label, ..callbacks) = physics.style(momentum-
arrows: true, show-edge-index: true)
 #draw(
   layout(g,beta: 100),
   source-style: source-style,
   sink-style: sink-style,
   edge-label: edge-label,
)
 #let g = parse(a.text).at(0)
 #draw(
   layout(layout(g,beta:
100,epochs:300,steps:200),subgraph:
subgraph.bits(g,
(false,false,false,false,false,true,true,true,true,true,true,true,true,true,true,true,true,true,true,true,true,true,true,tru
   source-style: source-style,
   sink-style: sink-style,
   edge-label: edge-label,
)
```
```


e0
e1
e2


e8
e7
e15
e16
e6
e13
e2
e11
e9
e5
e12
e8
e10
e14
e0
e4
e3

Parameters

```
style(
  map,
  default,
  typst-fields,
  scope,
  orientation-split,
  momentum-arrows,
  momentum-arrow-offset,
  momentum-arrow-length,
  momentum-arrow-ratio,
  momentum-arrow-stroke,
  momentum-arrow-mark,
  show-momentum,
  show-edge-index,
  show-half-edge-index,
  show-particle,
  momentum-fields,
  edge-index-fields,
  momentum-prefix,
  edge-index-prefix,
  half-edge-index-prefix,
  particle-prefix,
  label-separator,
  ],
  label-size,
  label-fill
) -> dictionary
```

massive `length`

Conventional massive-particle stroke thickness.

massless `length`

Conventional massless-particle stroke thickness.

dashed `array`

Dash pattern used for scalar-style lines.

dotted `string`

Dotted dash style used for ghost-style lines.

fermion-arrow-mark `dictionary`

CeTZ marker used for fermion particle-flow arrows on the main edge.

fermion-flow `dictionary`

Mark an edge-map entry as a fermion so the main edge receives one particle-flow arrow that follows `edge.orientation`. This is separate from optional momentum arrows.

wave `dictionary`

Photon-style wave pattern.

coil `dictionary`

Gluon-style coil pattern.

zigzag `dictionary`

Weak-boson-style zigzag pattern.

default-edge `dictionary`

Fallback style entry used when an edge has no known particle type.

default-map `dictionary`

Small built-in particle map for standalone physics diagrams. GammaLoop generated `edge-style.typ` files pass their model-specific map to `style`.

momentum-arrow-defaults `dictionary`

Default style for source-to-sink momentum arrow layers.

draw

- edge-halves()
- to-cetz-edge-halves()
- draw()

edge-halves

Split a laid-out graph edge into source and sink half-edge paths.

The returned dictionary has `source`, `sink`, and `curve`. The split point is the edge layout point, so the two half-edges join smoothly there.

Parameters

```
edge-halves(
  edge: dictionary,
  nodes: array,
  omega: float,
  source-outset: int float,
  sink-outset: int float,
  accuracy: float
) -> dictionary
```

edge `dictionary`

Edge record returned by `graph.edges(layout(g))`.

nodes `array`

Node records returned by `graph.nodes(layout(g))`.

omega `float`

Hobby curl used for the source-to-edge and edge-to-sink curves.

Default: `1.0`

source-outset `int` or `float`

Arc-length trim applied at the source node side.

Default: `0`

sink-outset `int` or `float`

Arc-length trim applied at the sink node side.

Default: `0`

accuracy `float`

Arc-length accuracy used while trimming.

Default: `0.001`

to-cetz-edge-halves

Draw the two halves of a laid-out graph edge through CeTZ.

`source-style` applies from the source node to the edge layout point, and `sink-style` applies from the edge layout point to the sink node.

Parameters

```
to-cetz-edge-halves(
  edge: dictionary,
  nodes: array,
  unit: int float length ratio,
  omega: float,
  source-outset: int float,
  sink-outset: int float,
  accuracy: float,
  source-style: dictionary,
  sink-style: dictionary
) -> content
```

edge `dictionary`

Edge record returned by `graph.edges(layout(g))`.

nodes `array`

Node records returned by `graph.nodes(layout(g))`.

unit `int` or `float` or `length` or `ratio`

Coordinate length for one graph-layout unit. Numbers are interpreted as em.

Default: `1`

omega `float`

Hobby curl used for the source-to-edge and edge-to-sink curves.

Default: `1.0`

source-outset `int` or `float`

Arc-length trim applied at the source node side.

Default: `0`

sink-outset `int` or `float`

Arc-length trim applied at the sink node side.

Default: `0`

accuracy `float`

Arc-length accuracy used while trimming.

Default: `0.001`

source-style `dictionary`

CeTZ style for the source half edge.

Default: `(:)`

sink-style `dictionary`

CeTZ style for the sink half edge.

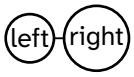
Default: `(:)`

draw

Draw a graph object with CeTZ.

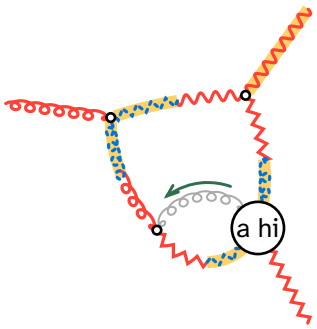
The graph must already have positions, either from `layout` or from explicit `pos` values passed to graph node or edge items. Paired edges without an explicit edge position use the midpoint of their endpoint nodes. Node and edge Typst style dictionaries or callbacks are evaluated and forwarded to CeTZ.

```
#let positioned = graph.build({
  graph.node(<left>, label: [left], pos: graph.pos(x: 0, y: 0))
  graph.node(<right>, label: [right], pos: graph.pos(ref: <left>, dx: 2.5, dy: 0))
  graph.edge(graph.source(<left>), graph.sink(<right>))
  graph.edge(graph.source(<right>), <right-out>, pos: graph.pos(ref: <right>, dx: 0.9, dy: 0.7))
},
  name: "positioned",
)
#draw(positioned, source-style: (stroke: black + 0.7pt), sink-style: (stroke: black + 0.7pt))
```




```
#let parallel-base-edge = 0
#let source-patterns = (
  "coil",
  "zigzag",
  "coil",
  "wave",
  "zigzag",
  "coil",
  "wave",
  "zigzag",
)
#let sink-patterns = (
  "coil",
  "zigzag",
  "coil",
  "zigzag",
  "coil",
  "wave",
  "coil",
  "wave",
)
#let parallel-layer(edge, mark: none) = if edge.eid == parallel-base-edge {
  (
    offset: -0.5,
    length: 1.5,
    ratio: 0.5,
    resolve-length: "min",
    stroke: (paint: rgb("#2f6f4e"), thickness: 1.1pt, cap: "round"),
    mark: mark,
  )
} else { none }
#let stack(base, layer) = if layer == none { base } else { (base, layer) }
```

```
#let source-stroke(edge) = if edge.eid == parallel-base-edge {
  (paint: gray, thickness: 0.75pt, cap: "round")
} else {
  (paint: red, thickness: 1.2pt, cap: "round")
}
#let sink-stroke(edge) = if edge.eid == parallel-base-edge {
  (paint: gray, thickness: 0.75pt, cap: "round")
} else {
  (paint: blue, thickness: 1.2pt, cap: "round", dash: "dotted")
}
#let source-style(edge) = {
  let base = (
    stroke: source-stroke(edge),
    pattern: source-patterns.at(edge.eid),
    pattern-amplitude: 0.18,
    pattern-wavelength: 0.55,
    pattern-coil-longitudinal-scale: 1.6,
  )
  stack(base, parallel-layer(edge))
}
#let sink-style(edge) = {
  let base = (
    stroke: sink-stroke(edge),
    pattern: sink-patterns.at(edge.eid),
    pattern-amplitude: 0.18,
    pattern-wavelength: 0.55,
    pattern-coil-longitudinal-scale: 1.6,
  )
  stack(base, parallel-layer(edge, mark: (end: (symbol: "straight"), scale: 0.75)))
}
#let g = graph.build({
  graph.node(<a>, label: [a hi])
  graph.node(<c>)
  graph.node(<d>)
  graph.node(<e>)
  graph.edge(graph.source(<a>), <ac>, graph.sink(<c>), pos: graph.pos(x: 0, y: 0.75, mode: "pin"))
  graph.edge(graph.source(<c>), <ca>, graph.sink(<a>, compass: "e"))
  graph.edge(graph.source(<c>), <cd>, graph.sink(<d>, compass: "e"))
  graph.edge(graph.source(<e>), <ed>, graph.sink(<d>, compass: "e"))
  graph.edge(graph.source(<e>), <ea>, graph.sink(<a>, compass: "e"))
  graph.edge(graph.source(<d>), <d-out>)
  graph.edge(graph.source(<e>, compass: "e"), <e-out>)
  graph.edge(graph.source(<a>), <a-out>)
},
  name: "demo",
)
#let layed-out = layout(g)
#let east = subgraph.compass(layed-out, "e")
#draw(layed-out, subgraph: east, source-style: source-style, sink-style: sink-style)
```

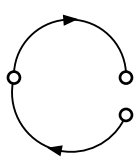



```
#let p = graph.build({
  graph.node(<pa>, label: [a], pos: graph.pos(y: 0, mode: "pin"))
```

```
  graph.node(<pc>, label: [c], pos: graph.pos(y: 0, mode: "pin"))
  graph.node(<pc1>, label: [c], pos: graph.pos(y: 0, mode: "pin"))
  graph.node(<pc2>, label: [c], pos: graph.pos(y: 0, mode: "pin"))
  graph.edge(graph.source(<pa>), <e0>, graph.sink(<pc>))
  graph.edge(graph.source(<pc2>), <e1>, graph.sink(<pc1>))
},
  name: "parallel demo",
)
#let parallel-edge-style(edge) = (
    offset: -0.5,
    length: 1.4,
    ratio: 0.5,
    resolve-length: "min",
    stroke: (paint: rgb("#2f6f4e"), thickness: 1.1pt, cap: "round"),
  )

#let focused-base-style(edge) = (
  stroke: (paint: gray, thickness: 0.7pt, cap: "round"),
  pattern: "coil",
  pattern-amplitude: 0.14,
  pattern-wavelength: 0.45,
  pattern-coil-longitudinal-scale: 1.5,
)
#let focused-source-style(edge) = (focused-base-style(edge), parallel-edge-style(edge))
#let focused-sink-style(edge) = (focused-base-style(edge), parallel-edge-style(edge) + (
  mark: (end: ">"),
))
#draw(layout(p, g-center: 0.004, label-steps: 0,), unit: 1.25, source-style: focused-source-style, sink-style: focused-sink-style)
```


```
#let g = graph.build({
  graph.node(<a>, pos: graph.pos(x: 0, y: 0, mode: "pin"))
  graph.node(<c>, pos: graph.pos(x: 3, y: 0, mode: "pin"))
  graph.node(<d>, pos: graph.pos(x: 3, y: -1, mode: "pin"))
  graph.edge(graph.source(<a>), <ac>, graph.sink(<c>), orientation: "default")
  graph.edge(graph.source(<a>), <ad>, graph.sink(<d>), orientation: "reversed")
},
  name: "oriented marks",
)
#let arrow = (
  end: (symbol: ">", fill: black, anchor: "center", shorten-to: auto),
  scale: 0.75,
)
#let oriented-arrow = (
  stroke: black + 0.7pt,
  mark: arrow,
  mark-position: "center-if-dangling",
  mark-orientation: "edge",
)
#draw(layout(g), unit: 1.4, source-style: oriented-arrow, sink-style: oriented-arrow)
```



Parameters

```
draw(
  graph: bytes,
  scope: dictionary,
  unit: int float length ratio,
  title: none auto content string,
  subgraph: none bytes,
  debug: bool int,
  node-radius: auto int float array,
  node-min-radius: int float,
  node-label-padding: int float,
  node-fill: any,
  node-stroke: any,
  node-outset: auto int float,
  node-label-style: dictionary,
  node-style: dictionary function none,
  node-label: auto content string function none,
  edge-stroke: any,
  edge-offset: int float,
  edge-length: none int float,
  edge-ratio: none int float,
  edge-resolve-length: string function,
  edge-accuracy: float,
  edge-optimize: bool,
  source-style: dictionary array function none,
  sink-style: dictionary array function none,
  edge-label: content string function none,
  edge-label-style: dictionary function none,
  edge-omega: float,
  edge-trim-accuracy: float,
  padding: none int float array dictionary,
  debug-edge-radius: int float,
  debug-edge-fill: any,
  debug-edge-stroke: any,
  debug-edge-label-fill: any,
  subgraph-edge-style: dictionary,
  subgraph-edge-underlay: bool
) -> content
```

graph `bytes`

Graph object with positions from `layout` or explicit graph API `pos` fields.

scope `dictionary`

Additional values merged into node and edge callback dictionaries.

Default: `(:)`

unit `int` or `float` or `length` or `ratio`

Coordinate length for one graph-layout unit. Numbers are interpreted as em.

Default: 1

title `none` or `auto` or `content` or `string`

Optional title displayed above the diagram. Use `auto` for the graph name.

Default: `none`

subgraph `none` or `bytes`

Optional subgraph whose half-edges are shaded.

Default: `none`

debug `bool` or `int`

Debug level. `1` enables CeTZ canvas debug; `2` also marks edge positions.

Default: `false`

node-radius `auto` or `int` or `float` or `array`

Default CeTZ node radius. Use `auto` to fit the node label.

Default: `auto`

node-min-radius `int` or `float`

Minimum radius used when `node-radius` is `auto`.

Default: `0.16`

node-label-padding `int` or `float`

Extra canvas-unit padding around labels when `node-radius` is `auto`.

Default: `0.08`

node-fill `any`

Default CeTZ node fill.

Default: `white`

node-stroke `any`

Default CeTZ node stroke.

Default: `black`

node-outset `auto` or `int` or `float`

Edge clearance from node centers. `auto` uses each node circle radius. Increase this when labels extend beyond the circle.

Default: `auto`

node-label-style `dictionary`

Default node-label style forwarded to `cetz.draw.content`.

Default: `(:)`

node-style `dictionary` or `function` or `none`

Node style dictionary or callback. A callback receives node data.

Default: `(:)`

node-label `auto` or `content` or `string` or `function` or `none`

Node label content or callback. `auto` uses the node name.

Default: `auto`

edge-stroke `any`

Default CeTZ edge stroke.

Default: `0.1em`

edge-offset `int` or `float`

Default normal offset for edge paths. Applied to the base edge geometry before patterns; node outsets then trim the shifted path.

Default: `0`

edge-length `none` or `int` or `float`

Maximum visible arc length for centered parallel edge paths. `none` keeps the full shifted path.

Default: `none`

edge-ratio `none` or `int` or `float`

Maximum visible fraction of the base edge length for centered parallel edge paths. Combined with `edge-length` according to `edge-resolve-length`.

Default: `none`

edge-resolve-length `string` or `function`

Resolve `edge-length` and `edge-ratio`. Accepted string values are `"min"`/`"shorter"`, `"max"`/`"longer"`, `"length"`/`"fixed"`, `"ratio"`/`"relative"`, or `"none"`/`"full"`. A function receives `(base-length, length, ratio)`.

Default: `"min"`

edge-accuracy `float`

Arc-length accuracy for fitted parallel edge paths.

Default: `0.001`

edge-optimize `bool`

Let Kurbo optimize the fitted parallel path.

Default: `true`

source-style `dictionary` or `array` or `function` or `none`

Source half-edge style dictionary, array of layer dictionaries, or callback. `mark-position: "center-if-dangling"` keeps an end marker at the paired-edge split point while centering it on dangling half edges. `mark-orientation: "edge"` makes a mark follow `edge.orientation` instead of raw path direction; reversed edges move the mark to the sink half and flip it, while undirected edges suppress it.

Default: `(:)`

sink-style `dictionary` or `array` or `function` or `none`

Sink half-edge style dictionary, array of layer dictionaries, or callback. A callback receives edge data. `mark-position: "center-if-dangling"` has the same dangling-edge behavior as for `source-style`, and `mark-orientation: "edge"` participates in the same orientation-aware mark placement.

Default: `(:)`

edge-label `content` or `string` or `function` or `none`

Edge label content or callback. A callback receives edge data.

Default: `none`

edge-label-style `dictionary` or `function` or `none`

Edge-label style forwarded to `cetz.draw.content`; use `anchor` to choose which point of the label is placed at the layout label position. A callback receives edge data.

Default: `(:)`

edge-omega `float`

Hobby curl used at the endpoints of paired edge curves.

Default: `1.0`

edge-trim-accuracy `float`

Arc-length accuracy for trimming edge curves at node outsets.

Default: `0.001`

padding `none` or `int` or `float` or `array` or `dictionary`

CeTZ canvas padding.

Default: `0.4`

debug-edge-radius `int` or `float`

Radius for edge-position markers shown at `debug >= 2`.

Default: `0.08`

debug-edge-fill `any`

Fill for edge-position markers shown at `debug >= 2`.

Default: `rgb("#ff9f1c")`

debug-edge-stroke `any`

Stroke for edge-position markers shown at `debug >= 2`.

Default: `rgb("#d72638") + 0.35pt`

debug-edge-label-fill `any`

Label fill for edge-position markers shown at `debug >= 2`.

Default: `rgb("#7a1020")`

subgraph-edge-style `dictionary`

CeTZ style used to shade half-edges included in `subgraph`.

Default: `(stroke: rgb("#ffd166") + 4.5pt)`

subgraph-edge-underlay `bool`

Draw subgraph shading below the normal half-edge style.

Default: `true`

subgraph

- label()
- bits()
- compass()
- to-label()
- hedges()
- contains()

label

Construct an subgraph object from a base62 label.

```
#let g = graph.build({
  graph.node(<a>)
  graph.node(<b>)
  graph.edge(graph.source(<a>, compass: "e"),
```

```
graph.sink(<b>))
})
#let east = subgraph.compass(g, "e")
#let same = subgraph.label(g, subgraph.to-
label(east))
#subgraph.hedges(same).len()
```

```
1
```

Parameters

```
label(
  graph: dictionary,
  label: string
) -> bytes
```

graph `dictionary`

Graph object whose half-edge set the label refers to.

label `string`

Base62 subgraph label returned by `to-label()` or produced by Linnest.

bits

Construct an subgraph object from a boolean hedge array.

```
#let g = graph.build({
  graph.node(<a>)
  graph.node(<b>)
  graph.edge(graph.source(<a>), graph.sink(<b>))
})
#let first = subgraph.bits(g, (true, false))
#subgraph.contains(first, 0)
```

```
true
```

Parameters

```
bits(
  graph: dictionary,
  bits: array
) -> bytes
```

graph `dictionary`

Graph object whose half-edge order defines the bit array.

bits `array`

Boolean array selecting half edges by graph half-edge index.

compass

Construct an subgraph object from a DOT compass point such as `"n"` or `"s"`.

```
#let g = graph.build({
  graph.node(<a>)
  graph.node(<b>)
  graph.edge(graph.source(<a>, compass: "e"),
graph.sink(<b>))
})
#subgraph.hedges(subgraph.compass(g, "e")).len()
```

```
1
```

Parameters

```
compass(
  graph: dictionary,
  compass: string
) -> bytes
```

graph `dictionary`

Graph object to filter by half-edge compass statement.

compass `string`

DOT compass point such as `"n"`, `"s"`, `"e"`, or `"w"`.

to-label

Convert an subgraph object to its base62 label.

```
#let g = graph.build({
  graph.node(<a>)
  graph.node(<b>)
  graph.edge(graph.source(<a>), graph.sink(<b>))
})
#subgraph.to-label(subgraph.bits(g, (true,
false)))
```

```
1
```

Parameters

```
to-label(subgraph: bytes) -> string
```

subgraph `bytes`

Subgraph object returned by this module or by `graph.cycles`/`graph.forests`.

hedges

Return the hedge indices included in an subgraph object.

```
#let g = graph.build({
  graph.node(<a>)
  graph.node(<b>)
  graph.edge(graph.source(<a>), graph.sink(<b>))
})
#subgraph.hedges(subgraph.bits(g, (true,
false)))
```

```
(0,)
```

Parameters

```
hedges(subgraph: bytes) -> array
```

subgraph `bytes`

Subgraph object to inspect.

contains

Test whether an subgraph object includes a hedge.

```
#let g = graph.build({
  graph.node(<a>)
  graph.node(<b>)
  graph.edge(graph.source(<a>), graph.sink(<b>))
})
#subgraph.contains(subgraph.bits(g, (true,
false)), 0)
```

```
true
```

Parameters

```
contains(
  subgraph: bytes,
  hedge: int
) -> bool
```

subgraph `bytes`

Subgraph object to inspect.

hedge `int`

Half-edge index to test.